

LAB 2

TECH 60 - Sec 14

F26

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Objective:

For Lab 2, a series circuit was constructed using resistors of various values. A DMM measured the resistance of each component, and a DC power supply set at 9V was sent through the circuit. Ohm's Law, as well as Kirchhoff's Voltage Law were utilized to calculate the total resistance, current, and to see if the source voltage and the voltage across the resistors agreed. Comparing the hand calculations, multisim, and breadboard, the data all agreed with each other, with the highest percent error being 2.53%.

Materials/Equipment:

- Digital Multimeter
- DC Power Supply
- 3 Resistors
- Solderless breadboard
- Wires
- Device with Multisim

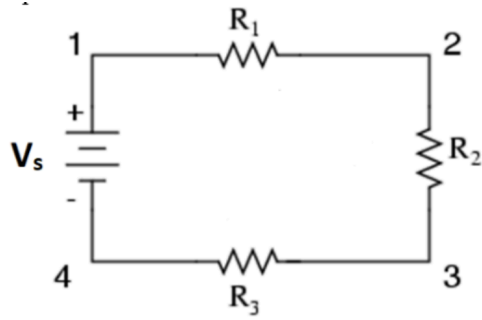
Data:

Part 1.

$$PE = \frac{\text{Nominal Value} - \text{Measured Value}}{\text{Nominal Value}} \times 100 \%$$

	R ₁ (3 kΩ)	R ₂ (2 kΩ)	R ₃ (1 kΩ)
1 st Band Color	orange	red	brown
2 nd Band Color	black	black	black
3 rd Band Color	red	red	red
4 th Band Color	gold	gold	gold
Nominal Value	3 kΩ ± 5%	2 kΩ ± 5%	1 kΩ ± 5%
Measured value	03.028 kΩ	02.009 kΩ	0.978 kΩ
%Error	0.93%	0.45%	2.2%

Part 2. Hand calculation of the circuit parameters



$V_s = 9V$, $R_1 = 3\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, and $R_3 = 1\text{ k}\Omega$, $R_T = 6\text{ k}\Omega$.

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 3\text{ k}\Omega + 2\text{ k}\Omega + 1\text{ k}\Omega$$

$$R_T = 6\text{ k}\Omega$$

$$I = V_s / R_T$$

$$I = 9V / 6\text{ k}\Omega$$

$$I = 1.5\text{ mA}$$

$$V_1 = I \cdot R_1 = (1.5\text{ mA})(3\text{ k}\Omega) = 4.5V$$

$$V_2 = I \cdot R_2 = (1.5\text{ mA})(2\text{ k}\Omega) = 3.0V$$

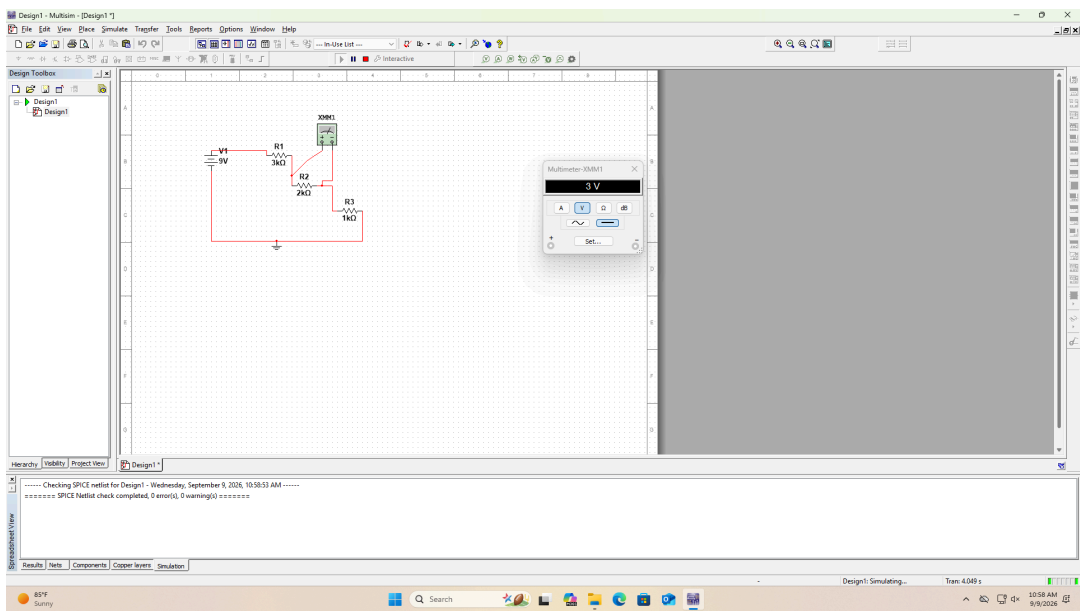
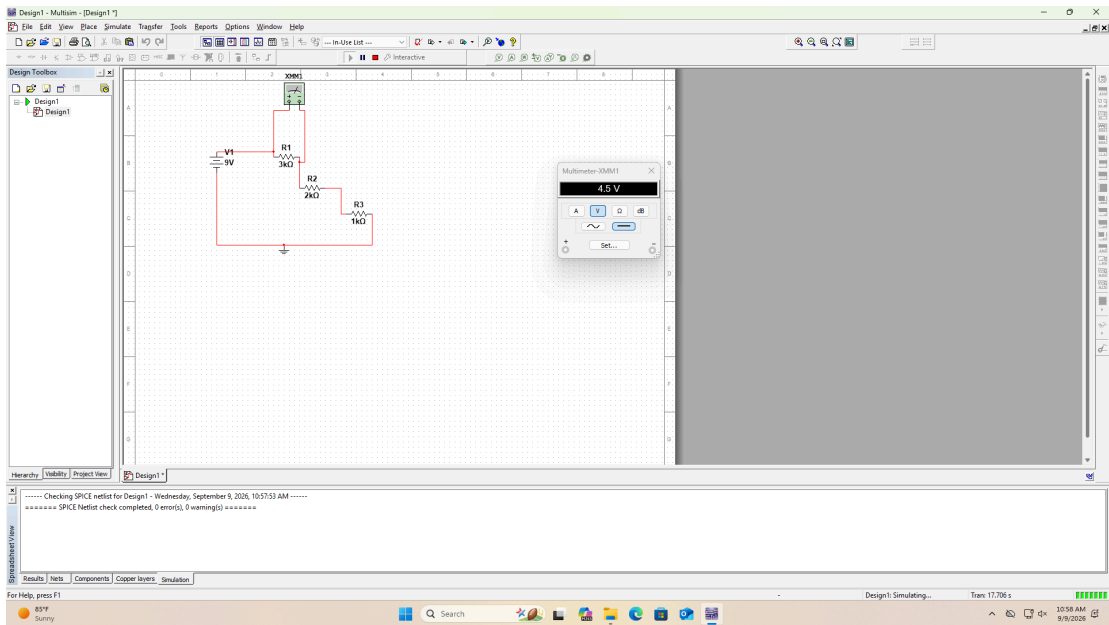
$$V_3 = I \cdot R_3 = (1.5\text{ mA})(1\text{ k}\Omega) = 1.5V$$

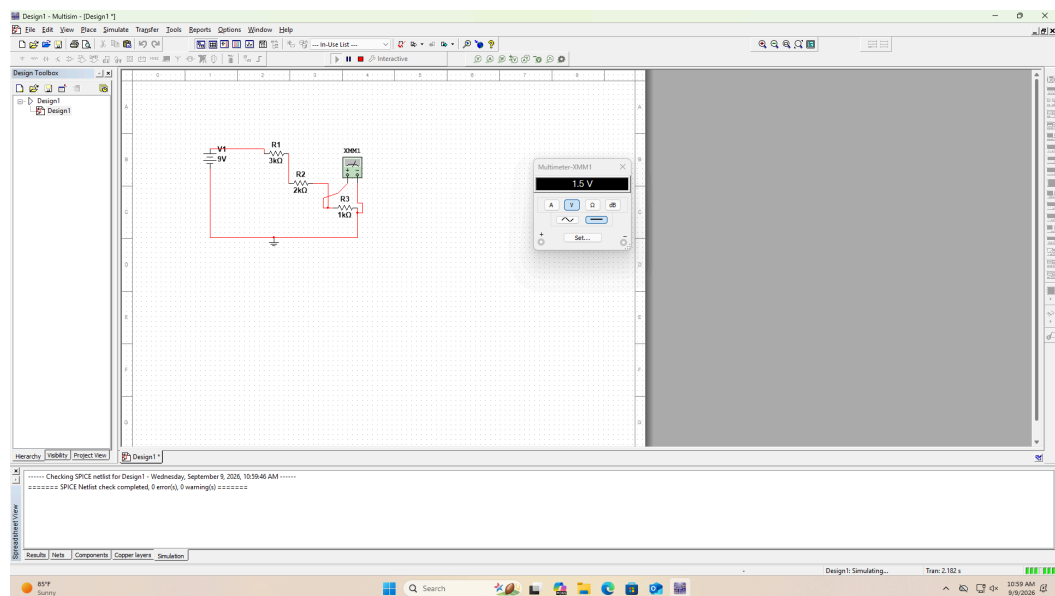
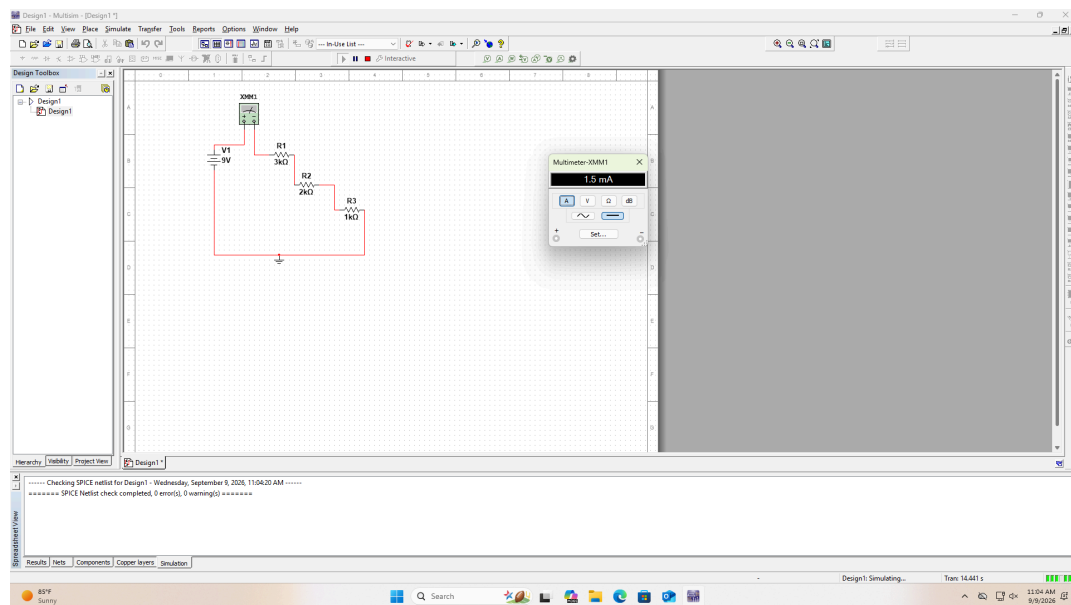
$$V_S = V_1 + V_2 + V_3$$

$$9V = 4.5V + 3.0V + 1.5V$$

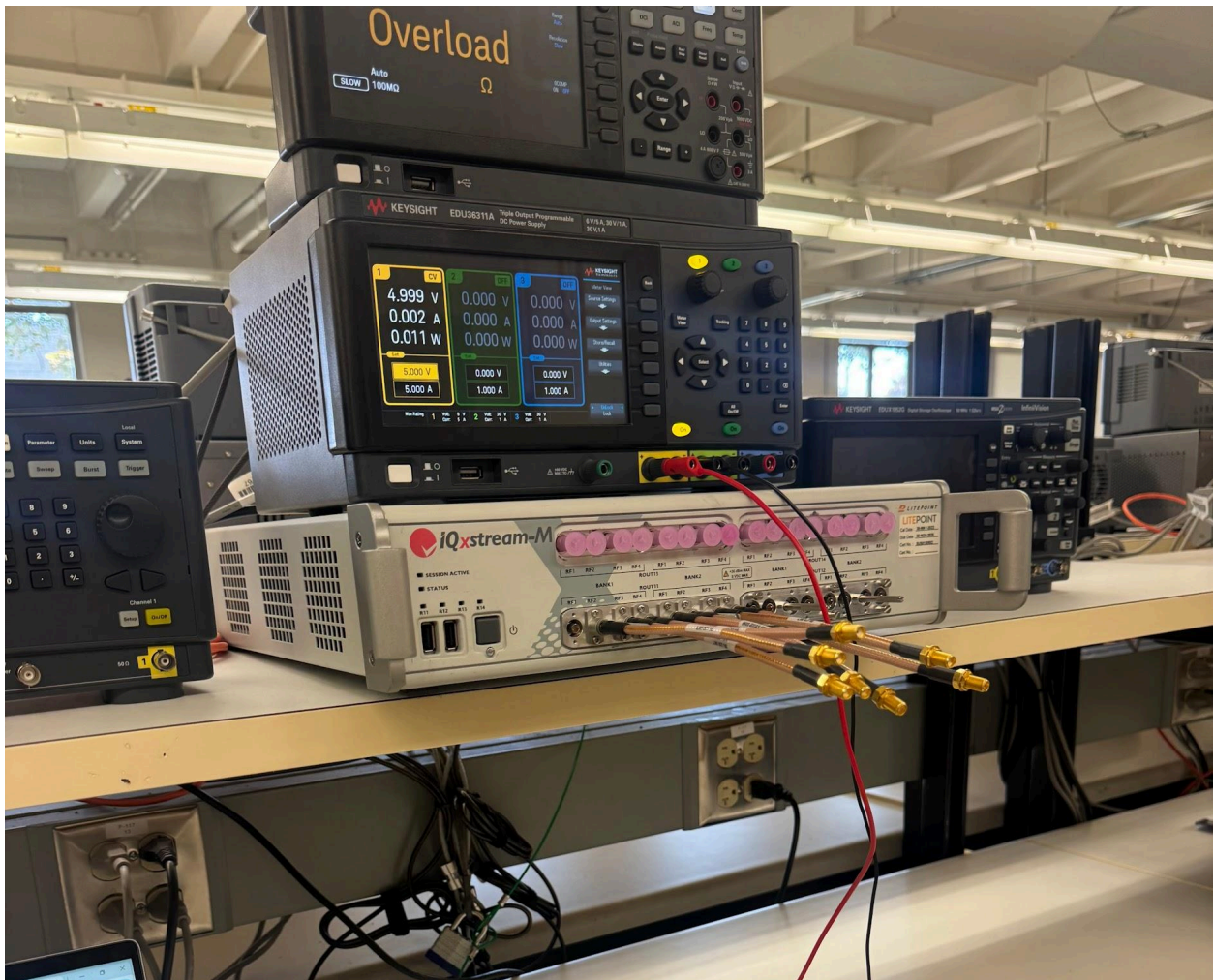
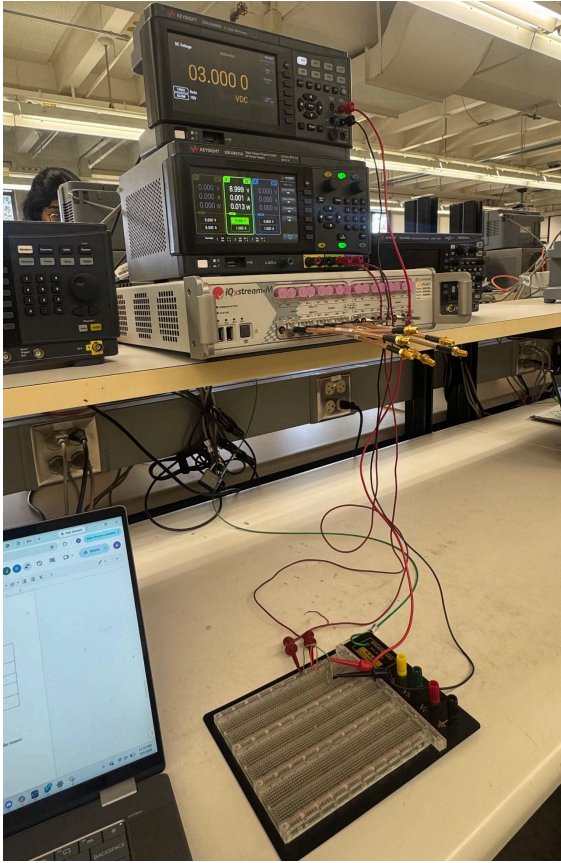
$$9V = 9V$$

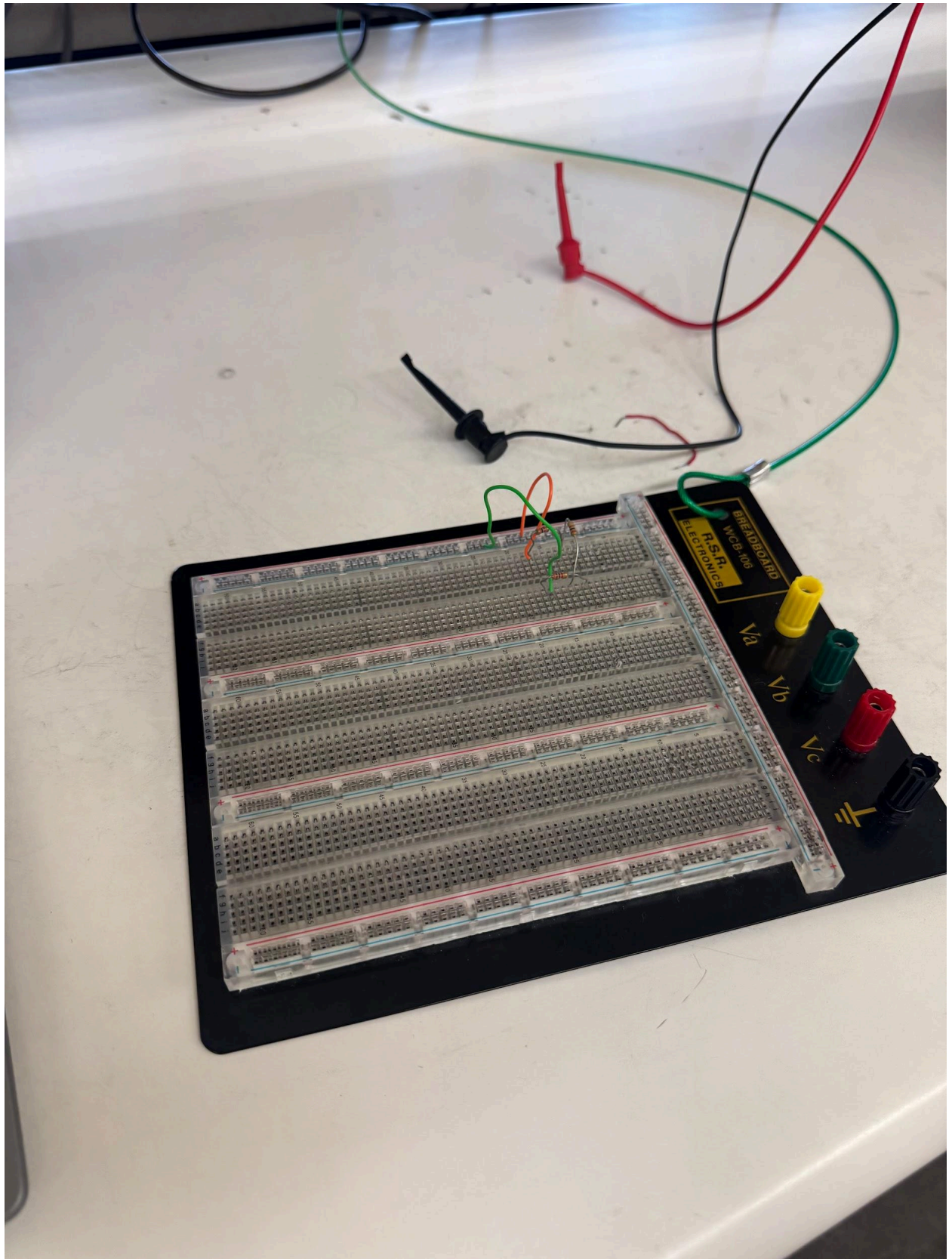
Part 3. Multisim Circuit Simulation





Part 4. Construct the circuit on the breadboard





Part 5.

Include all information in the table below:

	V_s	V_{R1}	V_{R2}	V_{R3}	$V_T = V_1 + V_2 + V_3$	%Difference between V_s and V_{Total}
Hand Calculation	9V	4.5V	3.0V	1.5V	$9V = 9V$	0%
Multisim	9V	4.5V	3.0V	1.5V	$9V = 9V$	0 %
Breadboard	8.998V	4.518V	3.0V	1.462V	$8.998 \approx 8.98$	0.2%
%Error	0.02%	0.4%	0%	2.53%	2.95% Total	Leave Blank

Questions

Analyze the above measurements to answer the following questions:

1: What do you observe is the relationship between V_{R1} , V_{R2} , V_{R3} in comparison to the source voltage. Write an expression and a few words to explain the relationship.

$$V_s = V_{R1} + V_{R2} + V_{R3}$$

The source voltage equals the sum of the voltages across each resistor in the circuit.

2. What do you notice is the relationship between the current through each resistor?
Write an expression or few words to explain the relationship.

The current depends on the total resistance from each resistor. If the resistance is lower, then the current goes up.

3. Does hand calculations, Multisim simulation, and breadboard measurements agree? Explain.

All 3 calculations did agree in this lab. For the hand calculations and the Multisim, we had a 0% percent difference. In addition, the breadboard circuit had a small difference of 0.2 %. The law held up in a real environment in almost an equal fashion as the simulated environment.

4. Does percent error remains under 5%. If the percent error is greater than 5% then explain the reason for the error

Percent error did remain under 5 %.

Challenges: The challenges that we had to face were trying to measure the volts because there were some errors when measuring the volts. It was mis-calculated and we had to change the wires a few times in order to get the measurements right. After making adjustments and making the voltages again, we were able to get the values that were close to our hand calculations and Multisim results.

Conclusion:

This lab had us build a simple circuit consisting of a 9V DC power source and three resistors of 3k, 2k, and 1k ohms respectively. We used Kirchhoff's Voltage Law (KVL): $V_S = V_1 + V_2 + V_3$ and confirmed it through 3 methods. We began with hand calculations to confirm the math. We then created the circuit on Multisim and on a physical breadboard to test the law in a simulation and a real environment. Overall, the results validated the theoretical relationships between total resistance, Ohm's Law, and the voltage divider rule.